NAME :- UMER SALEEM

CLASS :- BS CS (E)

ROLL NO :- 49

SUBMIT BY :- SIR ASAD

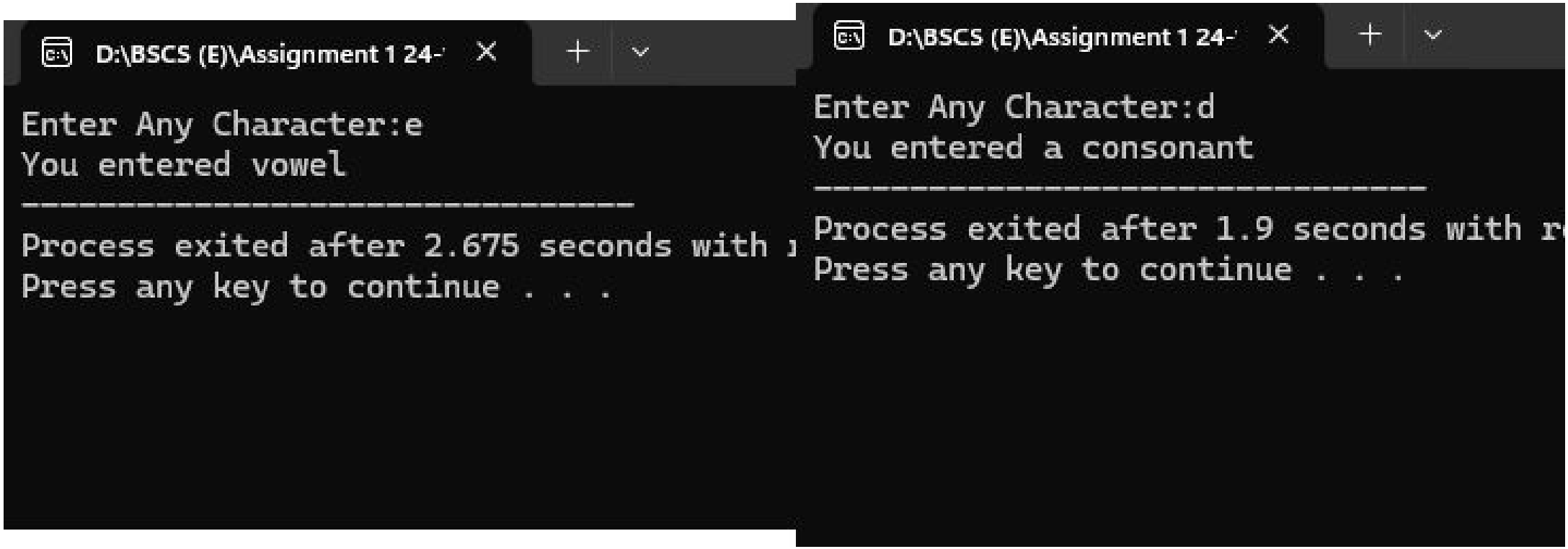
Q 1 *Write a program that reads an alphabet and determines whether the given alphabet is a vowel or consonant.*

Vowels or consonant in switch.cpp

```cpp
#include<iostream>
using namespace std;
int main()
{
    char alfabet;
    cout<<"Enter Any Character:";
    cin>>alfabet;
    switch(alfabet)
    {
        case 'a':
            cout<<"You entered vowel";
            break;
        case 'e':
            cout<<"You entered vowel";
            break;
        case 'i':
            cout<<"You entered vowel";
            break;
        case 'o':
            cout<<"You entered vowel";
            break;
        case 'u':
            cout<<"You entered vowel";
            break;
        default:
            cout<<"You entered a consonant";

    }
    return 0;
}
```

Output:-



Q 2 Write a program that reads a year and determines whether the given year is a leap year or not.

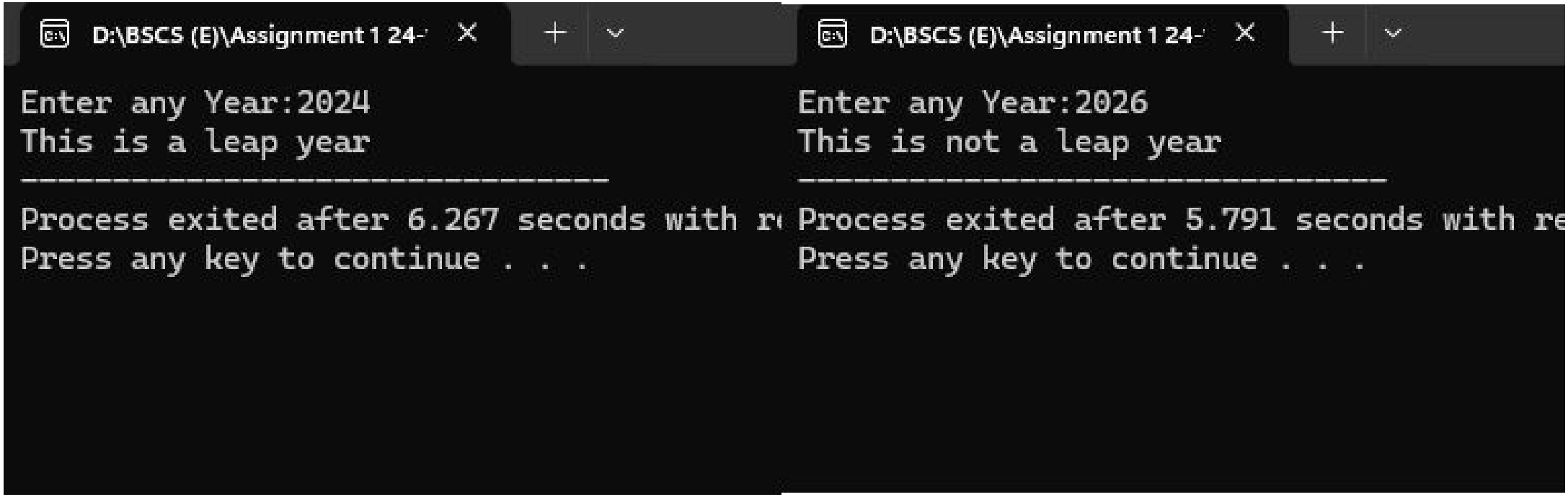
If Else Statment mein (Leap Year or Not).cpp

```cpp
//Write a program that reads a year and d
//not.


#include<iostream>
using namespace std;
int main()
{
    int year;
    cout<<"Enter any Year:";
    cin>>year;
    if(year%4==0)
    cout<<"This is a leap year";
    else
    cout<<"This is not a leap year";

    return 0;
}
```

Output:-



Q 3 Write a C++ program to construct an arithmetic calculator using if-else statements.

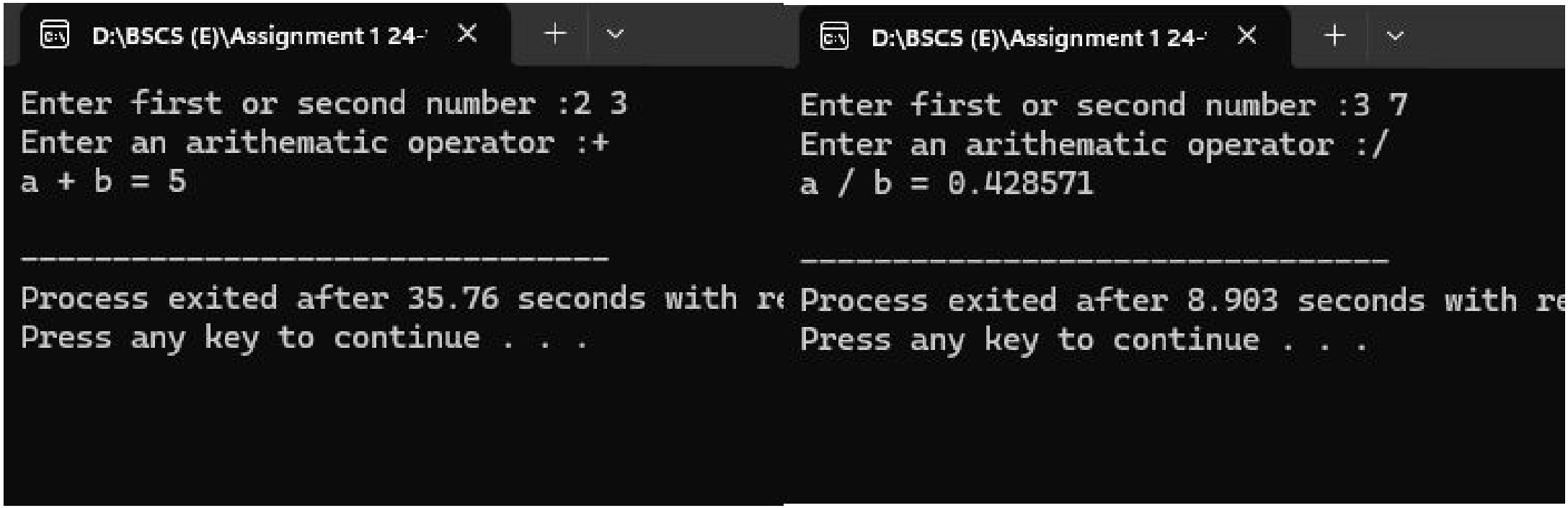
If Else mein(Calculater).cpp [*] Untitled2

```cpp
#include<iostream>
using namespace std;
int main()
{

    float a,b;
    char op;
    cout<<"Enter first or second number :";
    cin>>a>>b;
    cout<<"Enter an arithematic operator :";
    cin>>op;

    if(op == '+')
    cout<<"a + b = "<<a+b<<endl;
    else if(op=='-')
    cout<<"a - b = "<<a-b<<endl;
    else if(op=='*')
    cout<<"a * b = "<<a*b<<endl;
    else if(op=='/')
    {
    if(b==0)
    cout<<"Cannot divide by zero";
    else
    cout<<"a / b = "<<a/b<<endl;
    }
    else
    cout<<"You Entered Wrong Operator"<<endl;
    return 0;
}
```

Output:-



Q4. Write a program to swap two numbers entered by the user.

[*] Swipe Two Numbers.cpp

```cpp
//Write a program to swap two numbers e



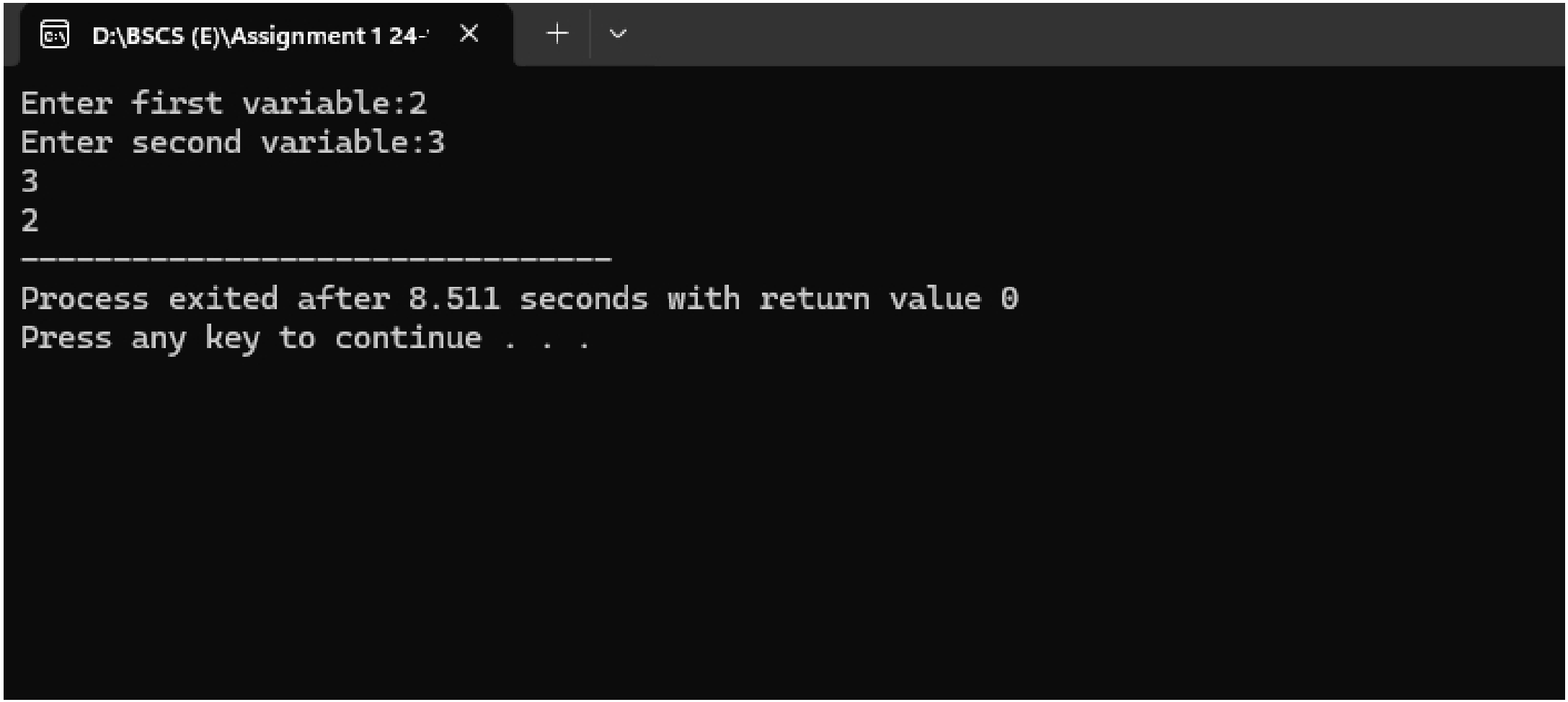
#include<iostream>
using namespace std;
int main()
{
    int var1,var2,temp;
    cout<<"Enter first variable:";
    cin>>var1;
    cout<<"Enter second variable:";
    cin>>var2;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout<<var1<<endl<<var2;

    return 0;
}
```

Output:-



Q5. Write a C++ program to find the third angle of a triangle.

Third Angle of Triangle.cpp

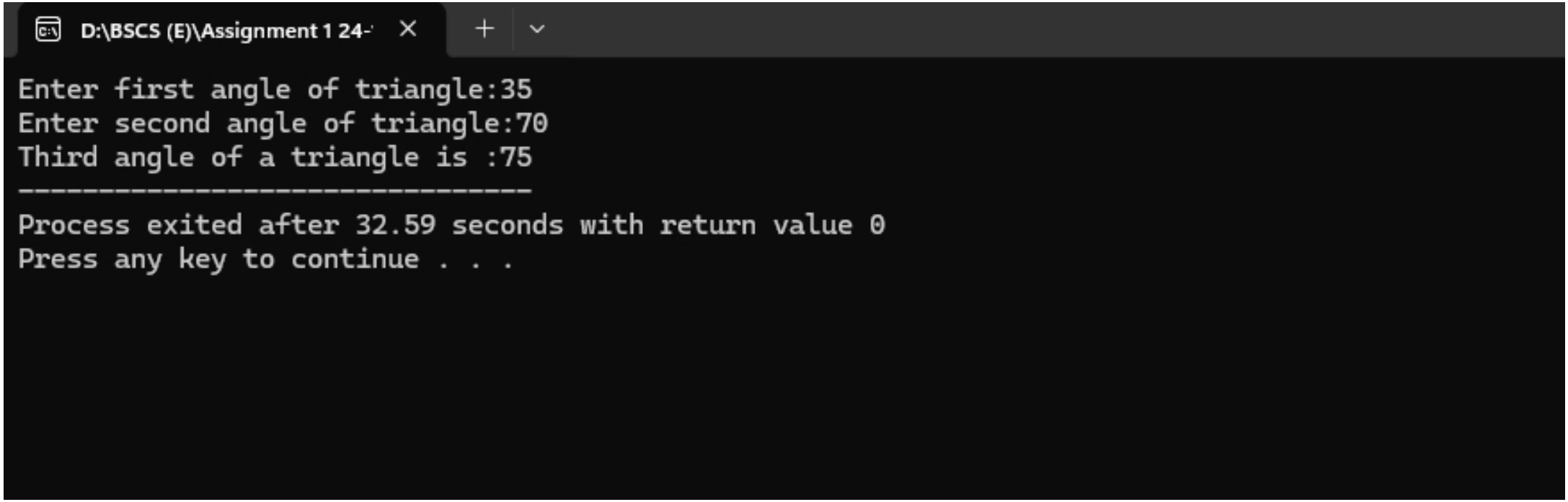
```cpp
// Write a C++ program to find the third angle of a tria

#include<iostream>
using namespace std;
int main()
{
    float a1,a2,a3;
    cout<<"Enter first angle of triangle:";
    cin>>a1;
    cout<<"Enter second angle of triangle:";
    cin>>a2;

    a3 = 180 - (a1 + a2);

    cout<<"Third angle of a triangle is :"<<a3;
    return 0;
}
```

Output:-



Q6. Write a C++ program to convert temperature in Celsius to Kelvin.

Celsius to Kelvin.cpp

```cpp
// Write a C++ program to convert temperature in C


#include<iostream>
using namespace std;
int main()
{
    float Celsius,Kelvin;

//  Take input from user:

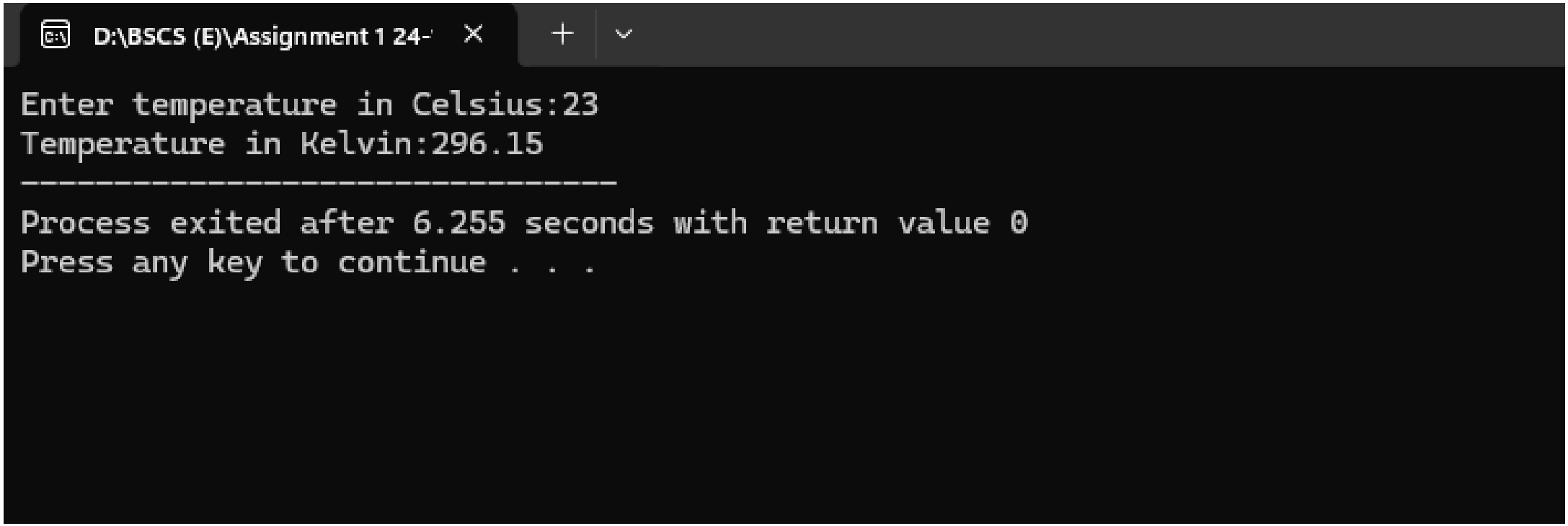
    cout<<"Enter temperature in Celsius:";
    cin>>Celsius;

//  Formula:

    Kelvin = Celsius + 273.15;

    cout<<"Temperature in Kelvin:"<<Kelvin;
    return 0;
}
```

Output:-



Q7. Write a C++ program to check whether a number is positive, negative, or zero.

Check Number is positve negative or zero.cpp

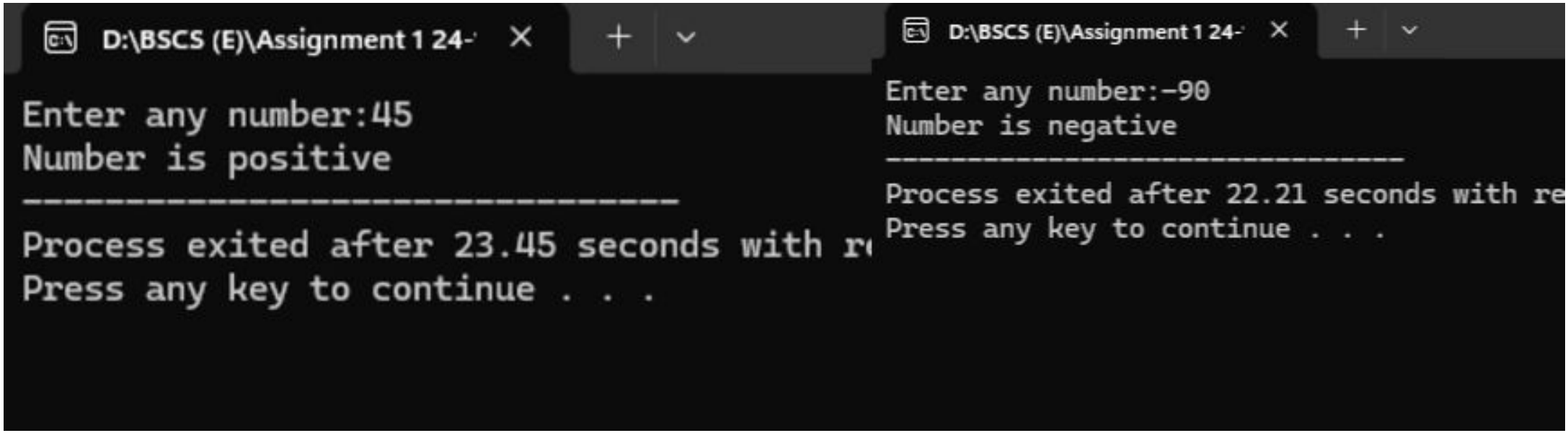
```cpp
//Write a C++ program to check whet

#include<iostream>
using namespace std;
int main()
{
    int num;
    cout<<"Enter any number:";
    cin>>num;

    if(num>0)
    cout<<"Number is positive";
    else if(num<0)
    cout<<"Number is negative";
    else
    cout<<"Number is zero";
    return 0;0

}
```

Output:-



Q8. Write a C++ program that accepts the user's first and last name and prints them in reverse order, with a space between them.

Swip First Name or Last Name.cpp

```cpp
//Write a C++ program that accepts the use
//order, with a space between them.

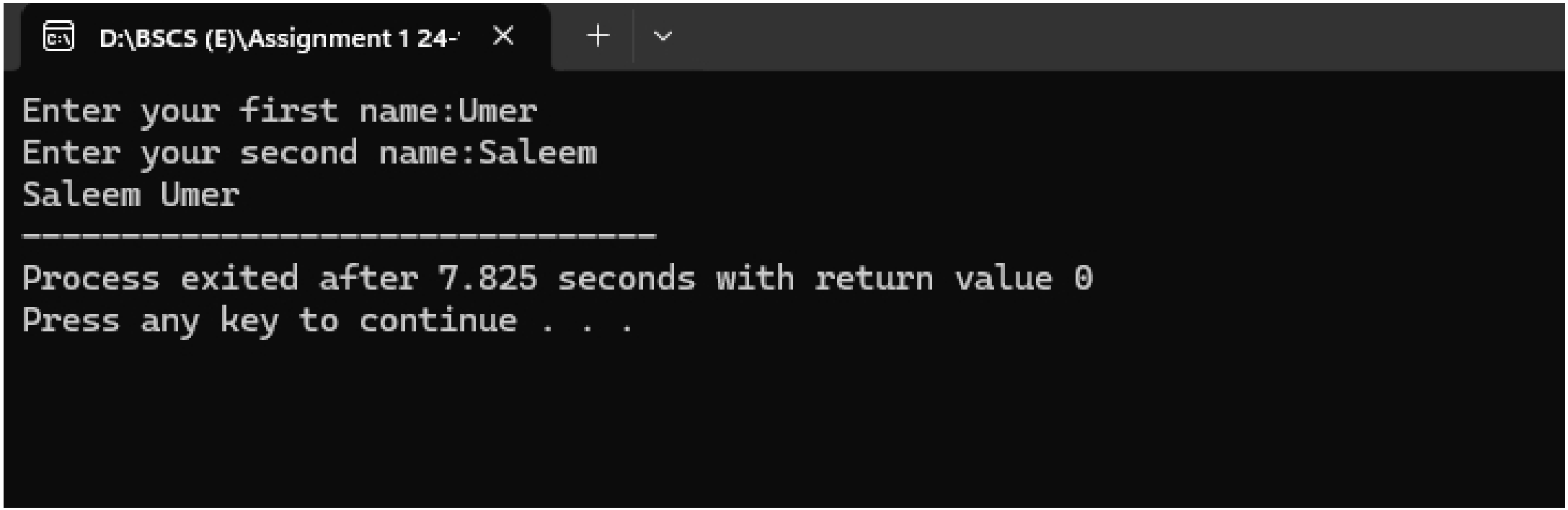

#include<iostream>
using namespace std;
int main()
{
    string First_Name,Last_Name;
    cout<<"Enter your first name:";
    cin>>First_Name;
    cout<<"Enter your second name:";
    cin>>Last_Name;

    cout<<Last_Name<<" "<< First_Name;
    return 0;


}
```

1 Sel: 0 Lines: 19 Length: 388 Insert Done parsing in 0.031 sec

Output:-



Q9. Write a C++ program to swap the values of two variables without using a third variable.

Swap two numeber without use third no.cpp

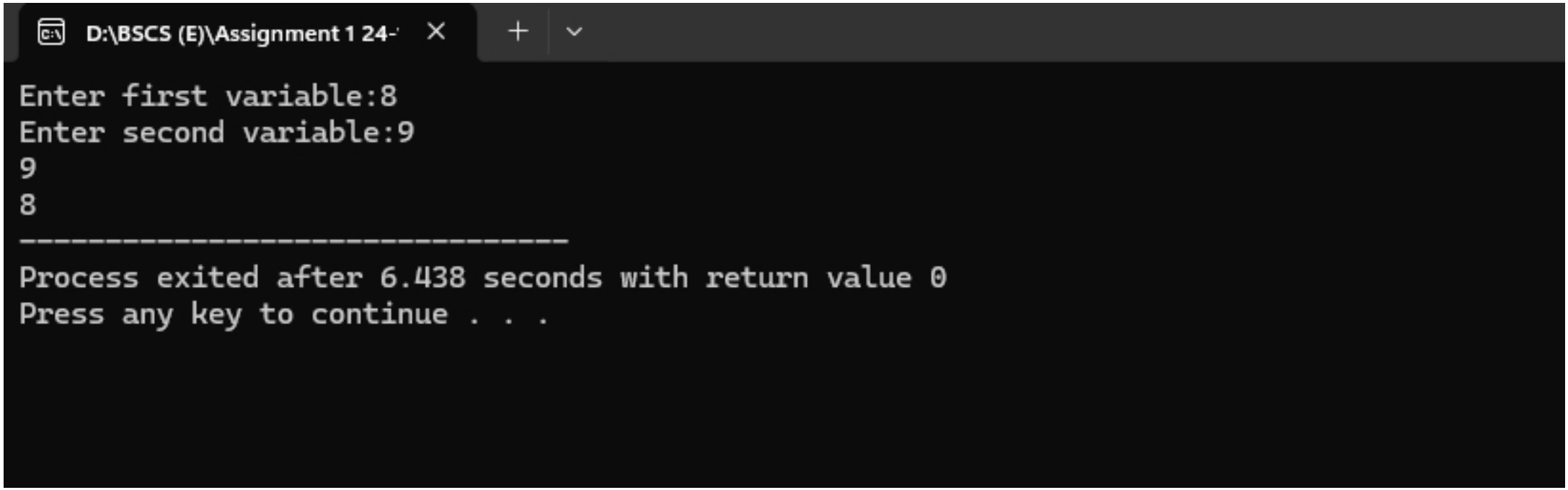
```cpp
//Write a C++ program to swap the value


#include<iostream>
using namespace std;
int main()
{
    int var1,var2;
    cout<<"Enter first variable:";
    cin>>var1;
    cout<<"Enter second variable:";
    cin>>var2;

    cout<<var2<<endl<<var1;
    return 0;

}
```

Output:-



Q10. Write a C++ program to print the code (ASCII code / Unicode code, etc.) of a given character.

Ascii Code of given character.cpp [*] Untitled3

```cpp
#include<iostream>
using namespace std;
int main()
{
    char ch;

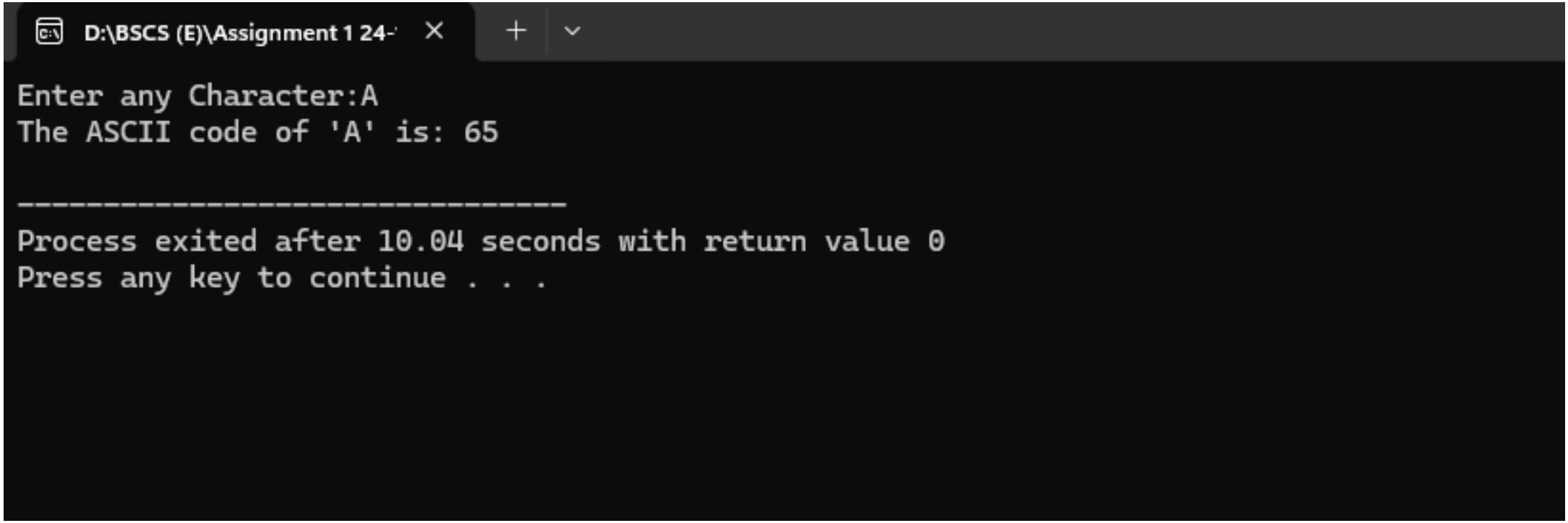
    cout<<"Enter any Character:";
    cin>>ch;

    cout << "The ASCII code of '" << ch << "' is: " << (int)ch << endl;

    return 0;
}
```

72 Sel: 0 Lines: 13 Length: 202 Insert Done parsing in 0.016 seconds

Output:-



Q11. Write C++ code to determine current, voltage, or resistance using Ohm's law.

Using Ohm Law To Calculate Voltage Current Resistance.cpp

```cpp
// Write C++ code to determine current, voltage, or resistance using Ohm'


#include<iostream>
using namespace std;
int main()
{
//  V = Voltage, I = Current , R = Resistance.
    char ch;
    double V,I,R;
    cout<<"Enter V for Volatge,I for current,R for Resistance:";
    cin>>ch;

    if((ch=='V')||(ch=='v'))
    {
        cout<<"Enter Current Value:";
        cin>>I;
        cout<<"Enter Resistance Value:";
        cin>>R;

        V=I*R;

        cout<<"Value of Voltage is :"<<V;
    }

    else if((ch=='I')||(ch=='i'))
    {
        cout<<"Enter Voltage Value:";
        cin>>V;
        cout<<"Enter Resistance Value:";
        cin>>R;

        I=V/R;

        cout<<"Value of Current is:"<<I;
    }
    else if((ch == 'R')||(ch=='r'))
    {
        cout<<"Enter Voltage Value:";
        cin>>V;
        cout<<"Enter Current Value:";
        cin>>I;

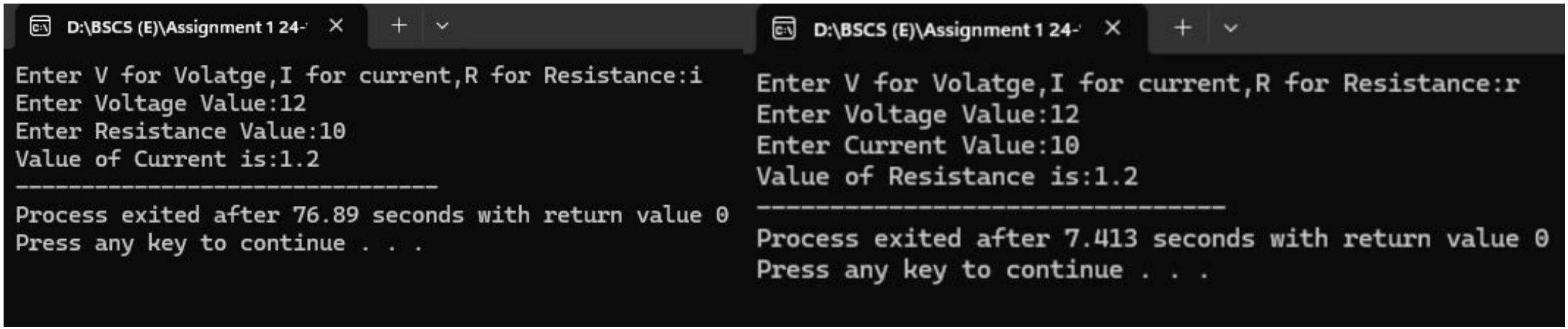
        R=V/I;

        cout<<"Value of Resistance is:"<<R;
    }

    else
    cout<<"Enter Wrong Charcter";

    return 0;
}
```

Output:-


D:\BSCS (E)\Assignment 1 24-
Enter V for Volatge,I for current,R for Resistance:i
Enter Voltage Value:12
Enter Resistance Value:10
Value of Current is:1.2

Process exited after 76.89 seconds with return value 0
Press any key to continue . . .
D:\BSCS (E)\Assignment 1 24-
Enter V for Volatge,I for current,R for Resistance:r
Enter Voltage Value:12
Enter Current Value:10
Value of Resistance is:1.2

Process exited after 7.413 seconds with return value 0
Press any key to continue . . .